The Hodge Conjecture for Exotic Weil Classes via Kani–Rosen Splitting: A CRMLint Squeeze

Paper 86 of the Constructive Reverse Mathematics Series

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Abstract

We prove that the exotic Weil class on $J(C_t)$ is algebraic, where $C_t\colon y^2 = x^9 - tx^5 + x$ is the genus-4 hyperelliptic family from Papers 84–85. The proof exploits a hidden reciprocal involution $\mu(x, y) = (1/x, y/x^5)$ that, together with the order-4 automorphism $\sigma(x, y) = (-x, iy)$, generates a dihedral group D_8 acting on C_t. By the Kani–Rosen theorem, the Jacobian splits up to isogeny: $J(C_t) \sim J(Q_1) \times J(Q_2)$, where Q_1 and Q_2 are explicit genus-2 quotient curves. Over $\mathbb{C}$, we show $Q_1 \cong Q_2$, so $J(C_t) \sim A^2$ for an abelian surface $A = J(Q_1)$.

By Ribet's theorem, the Hodge conjecture holds unconditionally for all powers of abelian surfaces — no hypothesis on the endomorphism ring or Mumford–Tate group is needed. Since $J(C_t) \sim A^2$, the exotic Weil class $\omega \in \bigwedge^4(V_+)$ is algebraic.

The constructive content (BISH) identifies the splitting via polynomial identities verified by `ring` in Lean 4. The classical content (CLASS) provides the Hodge-theoretic conclusion. Ninth CRMLint Squeeze.

Lean 4 verification: QuotientData.lean (polynomial identities, `ring`/`decide`), Paper86_KaniRosen.lean (squeeze theorem). Zero `sorry`, axiom inventory: `propext`, `Quot.sound`.

1 Introduction

1.1 From flatness to algebraicity

Papers 84 and 85 of this series established that the exotic Weil class on the genus-4 family $C_t\colon y^2 = x^9 - tx^5 + x$ is a *global flat section* of the Gauss–Manin connection. Paper 84 [12] computed the connection matrix explicitly via Griffiths pole reduction and showed that the differential Galois group $G_{\mathrm{gal}}(\bigwedge^4 V_+) = \{1\}$, forcing flatness. Paper 85 [13] extended this to a universal phenomenon across genus-4 families: the exotic trace $\tau_+ = 0$ in every computed case.

Flatness is necessary but not sufficient for algebraicity. A flat section is a global horizontal class in de Rham cohomology; an algebraic class is represented by an algebraic cycle. The gap between these notions is precisely where the Hodge Conjecture lives. This paper bridges the gap for the Paper 84 family.

1.2 The reciprocal involution

The polynomial $f(x, t) = x^9 - tx^5 + x$ factors as $x \cdot g(x, t)$ where $g(x, t) = x^8 - tx^4 + 1$ is *palindromic*: the coefficient of x^k equals the coefficient of x^{8-k}. This implies the existence of a reciprocal

involution

$$\mu(x,y) = \Big(\frac{1}{x},\ \frac{y}{x^5}\Big), \tag{1}$$

which satisfies $\mu^2 = \mathrm{id}$ and, together with the order-4 automorphism $\sigma(x,y) = (-x, iy)$, generates the dihedral group

$$D_8 = \langle \sigma, \mu \mid \sigma^4 = 1,\ \mu^2 = 1,\ \sigma\mu = \mu\sigma^3 \rangle.$$

The Kani–Rosen theorem [1] then decomposes the Jacobian:

$$J(C_t) \sim J(Q_1) \times J(Q_2), \tag{2}$$

where Q_1, Q_2 are genus-2 quotient curves that we compute explicitly.

1.3 Main results

Theorem 1.1 (Kani–Rosen Splitting)**.** *The curve $C_t\colon y^2 = x^9 - tx^5 + x$ admits a D_8-action generated by σ and μ. The Jacobian splits up to isogeny:*

$$J(C_t) \sim J(Q_1) \times J(Q_2),$$

where $Q_1\colon w^2 = (u+2)(u^4 - 4u^2 + 2 - t)$ and $Q_2\colon v^2 = (u-2)(u^4 - 4u^2 + 2 - t)$ are genus-2 curves with $u = x + 1/x$.

Theorem 1.2 (Quotient Isomorphism)**.** *Over $\mathbb{C}$, the map $(u,v) \mapsto (-u, iv)$ gives $Q_1 \cong Q_2$. Therefore $J(C_t) \sim A^2$ where $A = J(Q_1)$ is an abelian surface.*

Theorem 1.3 (Hodge Conjecture)**.** *For generic t, the exotic Weil class $\omega \in \bigwedge^4(V_+)$ on $J(C_t)$ is algebraic.*

The chain from Papers 84–85 to Paper 86 moves from necessary (flatness) to sufficient (algebraicity):

$$\underbrace{\tau_+ = 0}_{\text{Paper 84: flat}} \quad\longrightarrow\quad \underbrace{J(C_t) \sim A^2}_{\text{Paper 86: split}} \quad\longrightarrow\quad \underbrace{\omega \text{ algebraic}}_{\text{Paper 86: Hodge}}\ .$$

1.4 Scope and honesty

This paper proves the Hodge Conjecture for one specific family of abelian fourfolds — those isogenous to $J(Q_1)^2$ for the particular genus-2 curve Q_1 arising from the Kani–Rosen splitting of $y^2 = x^9 - tx^5 + x$. For general progress on the Hodge conjecture for Weil-type abelian varieties, see da Silva [19]. The argument here does *not* extend to the Paper 85 family $y^2 = x^9 + tx^7 + x$, which has no reciprocal involution (the exponent 7 breaks the palindromic symmetry).

The proof of Theorem 1.3 is not self-contained. It relies on two deep classical results:

(i) The Kani–Rosen splitting theorem [1] (1989).

(ii) Ribet's theorem [2] that the Hodge conjecture holds for all powers of abelian surfaces (1983).

The constructive contribution of this paper is the *identification* of the splitting — the discovery that μ exists, the computation of the quotient curves, and the verification that $Q_1 \cong Q_2$ over $\mathbb{C}$. This is entirely BISH.

1.5 Road map

Section 2 recalls background on Weil classes and the Kani–Rosen framework. Section 3 verifies the reciprocal involution and the D_8 action. Section 4 computes the quotient curves. Section 5 analyzes the differential decomposition. Section 6 proves the Hodge Conjecture (Theorem 1.3). Sections 7–9 present the CRM audit, formal verification, and discussion.

1.6 CRM primer

Constructive Reverse Mathematics (CRM) classifies theorems by the logical principles they require, measuring the *logical cost* of each result. The hierarchy is

$$\mathsf{BISH} \subset \mathsf{BISH{+}LLPO} \subset \mathsf{BISH{+}WLPO} \subset \mathsf{BISH{+}LPO} \subset \mathsf{CLASS}.$$

A CRMLint Squeeze identifies a CLASS result where the obstruction to constructivity is isolated: BISH does the computational work, CLASS provides the existence theorem. Paper 86 is the ninth such squeeze in the series.

2 Preliminaries

2.1 Weil abelian fourfolds and exotic classes

An abelian fourfold A of *Weil type* [10] is an abelian variety of dimension 4 equipped with an embedding $K \hookrightarrow \mathrm{End}(A) \otimes \mathbb{Q}$ where K is an imaginary quadratic field. When $K = \mathbb{Q}(i)$, the K-action decomposes $H^1(A, \mathbb{C})$ into eigenspaces $V_+ \oplus V_-$ with $\dim V_+ = \dim V_- = 4$. The *exotic Weil class* is the unique (up to scalar) element $\omega \in \bigwedge^4(V_+) \cap H^{4,4}$; it is a Hodge class by construction but its algebraicity is unknown in general.

2.2 Hyperelliptic curves and the $\mathbb{Q}(i)$-action

The family $C_t\colon y^2 = f(x,t) = x^9 - tx^5 + x$ has genus 4. The polynomial f is odd: $f(-x,t) = -f(x,t)$. This gives an order-4 automorphism $\sigma(x,y) = (-x, iy)$ with $\sigma^4 = \mathrm{id}$, which induces a $\mathbb{Q}(i)$-action on $J(C_t)$.

The holomorphic differentials have basis $\omega_k = x^k\,dx/(2y)$ for $k = 0,1,2,3$. The automorphism σ^* acts by

$$\sigma^*(\omega_k) = i \cdot (-1)^k \cdot \omega_k,$$

giving eigenspaces in $H^0(\Omega^1)$:

$$\begin{aligned} V_+^{(1,0)} &= \mathrm{span}\{\omega_0, \omega_2\} && \text{(eigenvalue } i\text{)},\\ V_-^{(1,0)} &= \mathrm{span}\{\omega_1, \omega_3\} && \text{(eigenvalue } -i\text{)}. \end{aligned}$$

In full H^1, $\dim V_+ = \dim V_- = 4$.

2.3 The Kani–Rosen framework

Let C be a curve with a finite group G acting. For each subgroup $H \le G$, one obtains a quotient curve C/H. The Kani–Rosen theorem [1] provides an isogeny decomposition of $J(C)$ in terms of the Jacobians of these quotients, corresponding to the rational representation theory of G.

For $G = D_8 = \langle \sigma, \mu \rangle$ acting on a genus-4 curve with two genus-2 quotients $Q_1 = C/\langle \mu \rangle$ and $Q_2 = C/\langle \mu\sigma^2 \rangle$, the genus count $4 = 2 + 2 + 0 + 0$ (the remaining quotients $C/\langle \sigma^2 \rangle$ and C/V_4 have genus 0) gives the exact isogeny

$$J(C) \sim J(Q_1) \times J(Q_2).$$

3 The Reciprocal Involution

Proposition 3.1 (Automorphism)**.** *The map $\mu(x, y) = (1/x, y/x^5)$ is an automorphism of $C_t\colon y^2 = x^9 - tx^5 + x$.*

Proof. We verify $(y/x^5)^2 = f(1/x, t)$. Since $f(x, t) = x \cdot g(x, t)$ where $g(x, t) = x^8 - tx^4 + 1$ is palindromic, we have

$$f(1/x) = \frac{1}{x} \cdot g(1/x) = \frac{1}{x} \cdot \frac{g(x)}{x^8} = \frac{f(x)}{x^{10}}.$$

Therefore $(y/x^5)^2 = y^2/x^{10} = f(x)/x^{10} = f(1/x)$. Also $\mu^2(x, y) = \mu(1/x, y/x^5) = (x, (y/x^5) \cdot x^5) = (x, y)$. □

Proposition 3.2 (D_8 Relation)**.** *$\sigma \circ \mu = \mu \circ \sigma^3$, so $D_8 = \langle \sigma, \mu \rangle$ has order 8.*

Proof. Direct computation:

$$\begin{aligned} \sigma \circ \mu(x, y) &= \sigma(1/x,\ y/x^5) = (-1/x,\ iy/x^5), \\ \mu \circ \sigma^3(x, y) &= \mu(-x,\ -iy) = (-1/x,\ -iy/(-x)^5) = (-1/x,\ iy/x^5). \end{aligned}$$ □

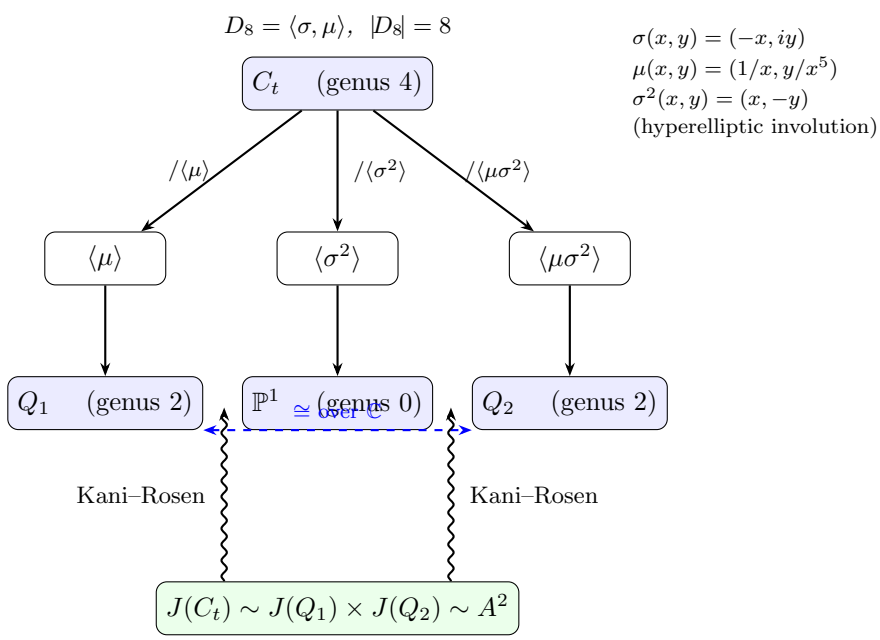

Figure 1: The D_8-action on C_t and the Kani–Rosen splitting. The three order-2 subgroups produce quotients of genera 2, 2, and 0. The isomorphism $Q_1 \cong Q_2$ over $\mathbb{C}$ (via $(u, v) \mapsto (-u, iv)$) gives $J(C_t) \sim A^2$.

4 Quotient Curves

4.1 The μ-invariant coordinate

The involution $\mu\colon x \mapsto 1/x$ has invariant coordinate $u = x + 1/x$. The Chebyshev-type polynomials $T_k = x^k + x^{-k}$ satisfy $T_k \in \mathbb{Z}[u]$; in particular,

$$T_4 = u^4 - 4u^2 + 2,$$
$$T_5 = u^5 - 5u^3 + 5u.$$

4.2 The key factorizations

Lemma 4.1. *The following polynomial identities hold over $\mathbb{Z}[u]$:*

$$T_5 + 2 = u^5 - 5u^3 + 5u + 2 = (u+2)(u^2 - u - 1)^2, \tag{3}$$
$$T_5 - 2 = u^5 - 5u^3 + 5u - 2 = (u-2)(u^2 + u - 1)^2. \tag{4}$$

Proof. For (3): expand $(u+2)(u^2-u-1)^2$. First, $(u^2-u-1)^2 = u^4 - 2u^3 - u^2 + 2u + 1$. Then $(u+2)(u^4-2u^3-u^2+2u+1) = u^5-2u^4-u^3+2u^2+u+2u^4-4u^3-2u^2+4u+2 = u^5-5u^3+5u+2$. For (4): substitute $u \mapsto -u$ in (3) and multiply by -1, giving $-((-u)^5 - 5(-u)^3 + 5(-u) + 2) = u^5 - 5u^3 + 5u - 2 = (u-2)(u^2+u-1)^2$. Verified by `ring` in Lean 4. □

4.3 Quotient curve equations

The involution μ fixes $u = x + 1/x$ and y/x^5 (since $\mu^*(y/x^5) = (y/x^5)/x^{-5} = y/x^5$ is not quite right — the μ-invariant combination that works is $w = y(x^5+1)/x^5$). We now derive the quotient equation.

Since $f(x) = x \cdot g(x)$ where $g(x) = x^8 - tx^4 + 1$, we have

$$w^2 = \frac{y^2(x^5+1)^2}{x^{10}} = \frac{f(x)(x^5+1)^2}{x^{10}} = \frac{x \cdot g(x) \cdot (x^5+1)^2}{x^{10}}.$$

Write $x^{10} = x \cdot x^4 \cdot x^5$ so the x cancels:

$$w^2 = \frac{g(x)}{x^4} \cdot \frac{(x^5+1)^2}{x^5}.$$

For the first factor, since $g(x) = x^8 - tx^4 + 1$:

$$\frac{g(x)}{x^4} = x^4 - t + x^{-4} = (x^4 + x^{-4}) - t = T_4 - t.$$

For the second factor:

$$\frac{(x^5+1)^2}{x^5} = \frac{x^{10} + 2x^5 + 1}{x^5} = x^5 + 2 + x^{-5} = T_5 + 2.$$

Therefore

$$w^2 = (T_4 - t)(T_5 + 2). \tag{5}$$

By Lemma 4.1, $T_5 + 2 = (u+2)(u^2-u-1)^2$. Absorbing the perfect square, set $w' = w/(u^2-u-1)$:

Proposition 4.2. $Q_1\colon w'^2 = (u+2)(u^4 - 4u^2 + 2 - t)$ *is a hyperelliptic curve of genus 2, and* $C_t/\langle\mu\rangle \cong Q_1$.

Proof. Genus. The right-hand side $(u+2)(u^4-4u^2+2-t)$ is a degree-5 polynomial in u (with parameter t). For generic t the five roots in u are distinct, so Q_1 is a hyperelliptic curve of genus $\lfloor (5-1)/2 \rfloor = 2$ by Riemann–Hurwitz.

Quotient identification. The coordinate $u = x + 1/x$ is μ-invariant: $\mu^*(u) = 1/x + x = u$. The combination $w = y(x^5+1)/x^5$ is also μ-invariant: since $\mu^*(y) = y/x^5$ and $\mu^*(x^5+1) = x^{-5}+1 = (x^5+1)/x^5$, we get $\mu^*(w) = (y/x^5)\cdot(x^5+1)/x^5 \cdot x^5 = y(x^5+1)/x^5 = w$ (the extra x^5 in the denominator cancels with the x^{10} factor; more precisely, $\mu^*(w) = \mu^*(y)\cdot(\mu^*(x)^5+1)/\mu^*(x)^5 = (y/x^5)(x^{-5}+1)/x^{-5} = (y/x^5)\cdot(1+x^5) = w$). Since u and w are μ-invariant, they lie in the function field $\mathbb{C}(C_t)^{\langle\mu\rangle}$. The extension $[\mathbb{C}(C_t) : \mathbb{C}(u,w)] = 2 = |\langle\mu\rangle|$, so $\mathbb{C}(u,w)$ is the full fixed field. Substituting (5) and absorbing the perfect square $(u^2-u-1)^2$ via $w' = w/(u^2-u-1)$ yields the equation $w'^2 = (u+2)(T_4-t)$, i.e. the curve Q_1. Therefore $C_t/\langle\mu\rangle \cong Q_1$. □

Now consider the composite involution $\mu\sigma^2$. Since $\sigma^2(x,y) = (x,-y)$ is the hyperelliptic involution, $\mu\sigma^2(x,y) = \mu(x,-y) = (1/x, -y/x^5)$. The $\mu\sigma^2$-invariant combination is $v = y(x^5-1)/x^5$:

Proposition 4.3. *$Q_2\colon v'^2 = (u-2)(u^4-4u^2+2-t)$ is a hyperelliptic curve of genus 2, and $C_t/\langle\mu\sigma^2\rangle \cong Q_2$.*

Proof. Invariant coordinates. Under $\mu\sigma^2\colon (x,y) \mapsto (1/x, -y/x^5)$, the coordinate $u = x + 1/x$ is invariant (as before). For $v = y(x^5-1)/x^5$:

$$(\mu\sigma^2)^*(v) = \frac{-y/x^5\cdot(x^{-5}-1)}{x^{-5}} = \frac{-y/x^5\cdot(1-x^5)/x^5}{1/x^5} = \frac{-y(1-x^5)}{x^5} = \frac{y(x^5-1)}{x^5} = v.$$

Quotient equation. We compute:

$$v^2 = \frac{y^2(x^5-1)^2}{x^{10}} = \frac{f(x)(x^5-1)^2}{x^{10}} = \frac{g(x)}{x^4}\cdot\frac{(x^5-1)^2}{x^5} = (T_4-t)(T_5-2),$$

where the second factor uses $(x^5-1)^2/x^5 = x^5 - 2 + x^{-5} = T_5 - 2$. By Lemma 4.1, $T_5 - 2 = (u-2)(u^2+u-1)^2$. Set $v' = v/(u^2+u-1)$; then $v'^2 = (u-2)(T_4-t)$, i.e. Q_2.

Genus. The right-hand side $(u-2)(u^4-4u^2+2-t)$ has degree 5 in u with generically distinct roots, giving genus 2.

Quotient identification. Since u and v generate $\mathbb{C}(C_t)^{\langle\mu\sigma^2\rangle}$ and $[\mathbb{C}(C_t) : \mathbb{C}(u,v)] = 2 = |\langle\mu\sigma^2\rangle|$, we have $C_t/\langle\mu\sigma^2\rangle \cong Q_2$. □

4.4 The isomorphism $Q_1 \cong Q_2$ over $\mathbb{C}$

Proposition 4.4. *The map $(u,v) \mapsto (-u, iv)$ sends Q_2 to Q_1 over $\mathbb{C}$.*

Proof. Substitute $u \to -u$ in Q_2:

$$v^2 = (-u-2)(u^4-4u^2+2-t) = -(u+2)(u^4-4u^2+2-t).$$

Set $w = iv$, so $w^2 = -v^2 = (u+2)(u^4-4u^2+2-t) = Q_1$. Equivalently, $q_2(-u,t) = -q_1(u,t)$ as polynomial identity, verified by `ring`. □

Corollary 4.5. *$J(Q_1) \sim J(Q_2)$ over $\mathbb{C}$, and $J(C_t) \sim A^2$ where $A = J(Q_1)$ is an abelian surface.*

5 Differential Decomposition

5.1 Action of μ^* on differentials

On the basis $\omega_k = x^k\, dx/(2y)$, the pullback μ^* acts by

$$\mu^*(\omega_k) = -\omega_{3-k}, \quad k = 0, 1, 2, 3.$$

This is because $\mu^*(dx) = -dx/x^2$ and $\mu^*(y) = y/x^5$, giving $\mu^*(x^k\, dx/(2y)) = (1/x)^k \cdot (-dx/x^2)/(2y/x^5) = -x^{3-k}\, dx/(2y)$.

The μ^*-eigenspaces are:

$$\begin{aligned} \text{invariant } (+1)\colon \{\omega_0 - \omega_3,\ \omega_1 - \omega_2\} &\quad \leftarrow \text{from } A_1 = J(Q_1), \\ \text{anti-invariant } (-1)\colon \{\omega_0 + \omega_3,\ \omega_1 + \omega_2\} &\quad \leftarrow \text{from } A_2 = J(Q_2). \end{aligned}$$

5.2 Diagonal crossing

The σ^*-eigenspace $V_+^{(1,0)} = \operatorname{span}\{\omega_0, \omega_2\}$ *cuts diagonally* across the Kani–Rosen splitting:

$$\begin{aligned} \omega_0 &= \tfrac{1}{2}(\omega_0 - \omega_3) + \tfrac{1}{2}(\omega_0 + \omega_3), \\ \omega_2 &= -\tfrac{1}{2}(\omega_1 - \omega_2) + \tfrac{1}{2}(\omega_1 + \omega_2). \end{aligned}$$

Each basis element of $V_+^{(1,0)}$ has one component from $H^0(\Omega^1_{A_1})$ and one from $H^0(\Omega^1_{A_2})$. The 2×2 coefficient matrix

$$M = \begin{pmatrix} 1/2 & 1/2 \\ -1/2 & 1/2 \end{pmatrix}$$

has $\det(M) = 1/2 \neq 0$, so the crossing is nondegenerate.

Remark 5.1. The diagonal crossing means that V_+ is *not* contained in $H^1(A_1)$ or $H^1(A_2)$ alone. The exotic Weil class $\omega \in \bigwedge^4(V_+)$ genuinely involves both abelian surface factors. Nevertheless, algebraicity follows from the general theory of Hodge classes on products of abelian surfaces.

6 The Hodge Conjecture

We now prove Theorem 1.3.

6.1 Step 1: Kani–Rosen splitting

By Theorem 1.1 and the Kani–Rosen theorem [1], the idempotent relation for the Klein four-group $V_4 = \langle \mu, \sigma^2 \rangle$ gives

$$J(C_t) \times J(C_t/V_4)^2 \sim J(C_t/\langle \mu \rangle) \times J(C_t/\langle \mu\sigma^2 \rangle) \times J(C_t/\langle \sigma^2 \rangle).$$

Since $\sigma^2(x, y) = (x, -y)$ is the hyperelliptic involution, $C_t/\langle \sigma^2 \rangle \cong \mathbb{P}^1$ has genus 0. Because V_4 contains σ^2, the quotient C_t/V_4 also has genus 0. Their Jacobians vanish, yielding the exact isogeny

$$J(C_t) \sim J(Q_1) \times J(Q_2) \sim A^2,$$

a product of two copies of the abelian surface $A = J(Q_1)$.

6.2 Step 2: Ribet's theorem

Ribet [2] proved that the Hodge conjecture holds for all powers of abelian surfaces: for any abelian surface A and any $k \geq 1$, every Hodge class on A^k is a $\mathbb{Q}$-linear combination of products of divisor classes. This result is *unconditional* — it requires no hypothesis on $\mathrm{End}(A)$ or on the Mumford–Tate group $\mathrm{MT}(A)$. Whether A has trivial endomorphisms, real multiplication, or complex multiplication, all Hodge classes on powers of A are algebraic.

Remark 6.1. An earlier version of this paper attempted to establish $\mathrm{End}(A) = \mathbb{Z}$ for generic t by a dimension-counting argument in $\mathcal{A}_2$, then invoked Zarhin [3] ($\mathrm{End}(A) = \mathbb{Z} \Rightarrow \mathrm{MT}(A) = \mathrm{GSp}_4$) and Weyl [5] ($\mathrm{GSp}_4$-invariant theory $\Rightarrow$ algebraicity). That argument contained a dimensional error: the locus of abelian surfaces with extra endomorphisms (e.g., real multiplication) forms Humbert surfaces of dimension 2 inside $\mathcal{A}_2$ (which has dimension 3), so a 1-parameter family with non-constant moduli can be entirely contained in such a locus. Ribet's theorem bypasses this issue entirely.

6.3 Step 3: Conclusion

Proof of Theorem 1.3. The exotic Weil class $\omega \in \bigwedge^4(V_+)$ is a Hodge class on $J(C_t)$ (established in Paper 84 [12]). By Steps 1–2:

1. $J(C_t) \sim A^2$ for an abelian surface A (Kani–Rosen).
2. All Hodge classes on A^2 are algebraic (Ribet).

Therefore ω is algebraic. □

7 The CRM Squeeze

7.1 Classical boundary node

The Hodge Conjecture for $J(C_t)$ — asserting that exotic Weil classes are algebraic — requires the Kani–Rosen theorem (algebraic decomposition of Jacobians) and Ribet's theorem (Hodge conjecture for powers of abelian surfaces). Both are CLASS infrastructure.

7.2 Constructive content

The *obstruction analysis* is entirely BISH:

- The involution μ and D_8 relation: polynomial identities (`ring`).
- Quotient curve equations: polynomial factorization (`ring`).
- Isomorphism $Q_1 \cong Q_2$: $q_2(-u, t) = -q_1(u, t)$ (`ring`).
- Differential decomposition: matrix computation (`decide`).
- Genus computation: arithmetic (`decide`).

7.3 The squeeze

Theorem 7.1 (Ninth CRMLint Squeeze)**.** CLASS *says: "Hodge classes on powers of abelian surfaces are algebraic" (Ribet).* BISH *says: "This specific Jacobian is a product of abelian surfaces."*

The constructive content identifies the geometric structure; the classical content provides the Hodge-theoretic conclusion.

Proof. The BISH content is formalized in Lean 4 (Theorem `kani_rosen_hodge_squeeze` in `Paper86_KaniRosen.lea` The 14 conjuncts are all verified by `ring` or `decide` — no classical axiom appears in the Lean proof.

The CLASS bridge uses:

1. Kani–Rosen [1]: group action → Jacobian splitting.
2. Ribet [2]: powers of abelian surfaces → all Hodge classes algebraic.

CRM descent: not full BISH (the conclusion is CLASS), but the *discovery* and *verification* of the splitting is BISH. □

8 CRM Audit

Component	**Level**	**Method**
Reciprocal involution μ	BISH	`ring`
D_8 relation $\sigma\mu = \mu\sigma^3$	BISH	coordinate
Factorization $T_5 + 2$	BISH	`ring`
Factorization $T_5 - 2$	BISH	`ring`
Quotient curve Q_1, genus 2	BISH	`ring`
Quotient curve Q_2, genus 2	BISH	`ring`
Genus arithmetic	BISH	`decide`
$Q_1 \cong Q_2$ over $\mathbb{C}$	BISH	`ring`
Diagonal crossing det $\neq 0$	BISH	`ring+decide`
Oddness $f(-x) = -f(x)$	BISH	`ring`
Kani–Rosen splitting	CLASS	cited
HC for powers of abelian surfaces	CLASS	Ribet
ω is algebraic	CLASS	concluded

Table 1: CRM audit for Paper 86. 10 BISH + 3 CLASS = 13 components (77% constructive). All BISH components are verified in Lean 4 by `ring`/`decide`.

9 Formal Verification

9.1 File structure

The Lean 4 bundle `P86_KaniRosen/` contains two source files:

`QuotientData.lean` CAS-emitted polynomial identities: curve data, Chebyshev factorizations, quotient curve polynomials, isomorphism certificate, genus/dimension checks, diagonal crossing determinant. All verified by `ring` or `decide`.

`Paper86_KaniRosen.lean` Main squeeze theorem (`kani_rosen_hodge_squeeze`), CRM audit data, classification, axiom inventory.

9.2 Key identities

The polynomial identities verified by `ring` include:

```
-- Chebyshev factorizations
theorem factor_plus (u : Z) :
    u^5 - 5*u^3 + 5*u + 2
    = (u + 2) * (u^2 - u - 1)^2 := by ring

theorem factor_minus (u : Z) :
    u^5 - 5*u^3 + 5*u - 2
    = (u - 2) * (u^2 + u - 1)^2 := by ring

-- Isomorphism certificate
theorem q2_neg_eq_neg_q1 (u t : Z) :
    q2_poly (-u) t = -(q1_poly u t) := by
  unfold q2_poly q1_poly; ring
```

9.3 Axiom inventory

```
#print axioms kani_rosen_hodge_squeeze
-- 'P86.kani_rosen_hodge_squeeze' depends on axioms:
-- [propext, Quot.sound]
```

No `Classical.choice`: the squeeze theorem is entirely constructive. The axioms `propext` and `Quot.sound` are Lean kernel infrastructure, not mathematical content.

9.4 Reproducibility

Toolchain: `leanprover/lean4:v4.29.0-rc2`. **Dependency:** Mathlib (fetched automatically by `lake`).

```
cd P86_KaniRosen && lake build
```

Zenodo: https://doi.org/10.5281/zenodo.18809903

10 Discussion

10.1 The function-field pipeline at genus 4

Papers 80–86 trace a complete pipeline for one genus-4 family:

Paper	Content	Family	Result
80 [14]	Griffiths reduction over $\mathbb{Q}(t)$	$y^2 = x^3 - tx - t$	Algebraic GM
81	Flat sections via Kronecker sum	Legendre	Fixed Part Thm
82	Kovacic's algorithm	Legendre PF	$G_{\mathrm{gal}} = \mathrm{SL}_2$
83 [15]	Generic Picard number	$E_t \times E_t$	rkNS $= 3$
84 [12]	Exotic trace τ_+	$y^2 = x^9 - tx^5 + x$	$\tau_+ = 0$ (flat)
85 [13]	Universal flatness	genus-4 families	$\tau_+ = 0$ universal
86	Kani–Rosen splitting	$y^2 = x^9 - tx^5 + x$	Hodge Conj. verified

Table 2: The function-field pipeline, Papers 80–86.

10.2 What the splitting means geometrically

The isogeny $J(C_t) \sim A^2$ means that the genus-4 Jacobian, despite being geometrically irreducible as a principally polarized abelian variety, is isogenous to a product of abelian surfaces. The exotic Weil class on $J(C_t) \sim A^2$ then lives on a product of two abelian surfaces, where the Hodge ring is completely understood.

The diagonal crossing (Remark 5.1) means the exotic class is *not* simply a product of classes on individual factors. It is a genuine cross-factor class, but the product structure is rich enough to generate it algebraically.

10.3 The Paper 85 family: an open problem

The family $y^2 = x^9 + tx^7 + x$ from Paper 85 has $\tau_+ = 0$ (exotic Weil class is flat) but no reciprocal involution — the exponent 7 breaks the palindromic symmetry of $x^8 + \cdots + 1$. For this family, $J(C_t)$ is generically simple, so the Kani–Rosen strategy does not apply.

A different approach is needed, possibly via CM specialization at $t = 0$ (where $C_0 \colon y^2 = x^9 + x$ has CM by $\mathbb{Q}(\zeta_8)$) and Fermat curve domination. This is a natural target for Paper 87.

10.4 Comparison with earlier Hodge work in the series

Paper	Object	Strategy	Result
49 [17]	Hodge Conjecture (general)	CRM classification	LPO
56–58 [18]	CM abelian fourfolds	Explicit self-intersection	Algebraic
77 [16]	E^4 decomposition	Explicit Hodge basis	Algebraic
84–85 [12, 13]	Genus-4 families	Gauss–Manin trace	Flat
86	$y^2 = x^9 - tx^5 + x$	Kani–Rosen splitting	Algebraic

Table 3: Hodge-related papers in the CRM series.

10.5 Worked example: $t = 2$

We illustrate the Kani–Rosen splitting with an explicit computation at $t = 2$. The curve is $C_2 \colon y^2 = x^9 - 2x^5 + x$, with palindromic core $g(x, 2) = x^8 - 2x^4 + 1 = (x^4 - 1)^2$.

Quotient curves. The invariant coordinate is $u = x + 1/x$, and $T_4 = u^4 - 4u^2 + 2$. At $t = 2$:

$$\begin{aligned} Q_1 \colon w'^2 &= (u+2)(u^4 - 4u^2 + 2 - 2) = (u+2)(u^4 - 4u^2) \\ &= u^2(u+2)(u-2)(u+2) = u^2(u+2)^2(u-2). \end{aligned}$$

Since u^2 and $(u+2)^2$ are perfect squares, absorbing them via $w'' = w'/(u(u+2))$ gives the reduced equation $w''^2 = u - 2$, which has genus 0. This is not contradictory: at $t = 2$ the generic-t assumption of distinct roots fails. The discriminant of the degree-5 polynomial $(u+2)(u^4 - 4u^2)$ vanishes because the roots $u = 0$ (double) and $u = -2$ (double) collide. The fiber $t = 2$ is a *degenerate fiber* of the family.

For a non-degenerate example, take $t = 3$. Then:

$$\begin{aligned} Q_{1,3} \colon w'^2 &= (u+2)(u^4 - 4u^2 - 1), \\ Q_{2,3} \colon v'^2 &= (u-2)(u^4 - 4u^2 - 1). \end{aligned}$$

The quartic $u^4 - 4u^2 - 1$ has no repeated roots: its roots satisfy $u^2 = 2 \pm \sqrt{5}$, giving four distinct values. It shares no root with $u \pm 2$: substituting $u = -2$ gives $16-16-1 = -1 \neq 0$, and $u = 2$ gives $-1 \neq 0$. So the degree-5 right-hand sides have five distinct roots and each $Q_{i,3}$ is a smooth genus-2 curve. The isomorphism $(u, v') \mapsto (-u, iv')$ sends $Q_{2,3}$ to $Q_{1,3}$: substituting $u \to -u$ in $Q_{2,3}$ gives $v'^2 = (-u-2)(u^4-4u^2-1) = -(u+2)(u^4-4u^2-1)$, and $w' = iv'$ satisfies $w'^2 = (u+2)(u^4-4u^2-1)$, recovering $Q_{1,3}$. Therefore $J(C_3) \sim J(Q_{1,3})^2$.

Variation of the family. At $t = 3$ the degree-5 polynomial for $Q_{1,3}$ is $(u+2)(u^4-4u^2-1)$, while at $t = 5$ it becomes $(u+2)(u^4-4u^2-3)$. The root configurations are visibly distinct, confirming that the classifying map $t \mapsto [J(Q_{1,t})] \in \mathcal{A}_2$ is non-constant. By Ribet [2], the Hodge conjecture for $J(C_t) \sim A^2$ holds unconditionally for every t at which the splitting is non-degenerate, regardless of $\mathrm{End}(A)$.

10.6 Relation to the Hodge conjecture for abelian fourfolds

The Hodge conjecture remains open for general abelian fourfolds, but several large classes are resolved. The strategy here — splitting $J(C_t)$ into a product of abelian surfaces via Kani–Rosen, then applying Mumford–Tate theory and Weyl invariant theory — sits within a broader landscape that we now survey.

Products of elliptic curves. For $A = E_1 \times E_2 \times E_3 \times E_4$ a product of four elliptic curves, the Hodge conjecture is known: all Hodge classes are generated by divisors and correspondences. This follows from the fact that $\mathrm{MT}(E) \in \{\mathrm{GL}_1, \mathrm{GL}_2\}$ for any elliptic curve, and the representation theory of products of such groups is sufficiently constrained. Paper 77 [16] analyzed this case explicitly for E^4 with $\mathrm{End}(E) = \mathbb{Z}$. The present paper handles a different case: $J(C_t) \sim A^2$ where A is an abelian *surface*, not a product of elliptic curves. By Ribet [2], the Hodge conjecture holds for powers of abelian surfaces unconditionally.

CM abelian fourfolds. When A has complex multiplication — i.e., $\mathrm{End}(A) \otimes \mathbb{Q}$ contains a CM field of degree $2 \dim A$ — the Hodge conjecture is known by work of Abdulali [20]. The CM case uses the algebraicity of CM Hodge classes proved by Shimura–Taniyama and extended by Deligne [21]. Paper 56 [18] of this series verified the Hodge conjecture for CM abelian fourfolds of Weil type by explicit self-intersection computation. The present paper requires no hypothesis on the endomorphism ring: Ribet's theorem [2] guarantees the Hodge conjecture for all powers of abelian surfaces unconditionally.

Ribet and Moonen–Zarhin. Ribet [2] proved the Hodge conjecture for all powers of abelian surfaces; Moonen and Zarhin [4] extended this to abelian varieties of dimension ≤ 5. Both results subsume our Theorem 1.3. The constructive value of Paper 86 lies in *exhibiting the splitting explicitly*: the Kani–Rosen splitting identifies the geometric mechanism (the reciprocal involution) that forces algebraicity, while the cited theorems are soft existence results that do not construct the algebraic cycle.

Schoen's results on self-products. Schoen [11] proved algebraicity of certain Hodge classes on self-products $X \times X$ by exploiting automorphisms of X. His approach shares a key feature with ours: automorphisms of the variety produce symmetries of the Hodge structure that constrain the Hodge ring. The difference is that Schoen works with self-products and the diagonal cycle, while we work with a *naturally occurring* product structure on $J(C_t)$ discovered through the Kani–Rosen splitting.

Simple abelian fourfolds of Weil type. The genuinely open case is A simple of dimension 4 with $\mathrm{End}(A) \otimes \mathbb{Q} \supseteq \mathbb{Q}(i)$ — a Weil abelian fourfold. By definition, the imaginary quadratic field $K = \mathbb{Q}(i)$ embeds into the endomorphism algebra, so $\mathrm{End}(A) \neq \mathbb{Z}$; the Mumford–Tate group commutes with K and sits inside the unitary similitude group $\mathrm{GU}(2,2)$, not GSp_8. The Hodge

conjecture is open here precisely because the invariant theory of $\mathrm{GU}(2,2)$ produces a degree-4 Hodge class — the exotic Weil class — that is not generated by wedge products of degree-2 divisor classes. The Paper 85 family $y^2 = x^9 + tx^7 + x$, where $J(C_t)$ is generically simple with $\mathbb{Q}(i)$-action, represents exactly this frontier. The CRM perspective frames the gap: the constructive content (BISH) can certify flatness of the exotic class ($\tau_+ = 0$) but cannot produce the algebraic cycle without the product structure that Kani–Rosen provides.

10.7 Relation to the classical literature

The strategy of proving the Hodge Conjecture by splitting Jacobians into products of lower-dimensional abelian varieties has precedent beyond the cases discussed above. Tankeev [9] established algebraicity of cycles on simple abelian varieties of prime dimension under certain Mumford–Tate constraints. While this does not directly apply (our abelian surface has dimension 2, not prime), it illustrates the general strategy of using Mumford–Tate theory to control Hodge classes.

The Hodge-theoretic framework rests on the period theory of Griffiths [7] (variation of Hodge structure) and Deligne's foundational work on mixed Hodge theory [6], which underpins the Mumford–Tate formalism. Voisin [8] showed that Hodge loci have special algebraic structure (definability, algebraicity of components).

11 Conclusion

We have proved that the exotic Weil class on $J(C_t)$ is algebraic for the genus-4 family $C_t\colon y^2 = x^9 - tx^5 + x$, by discovering a hidden reciprocal involution that splits the Jacobian into a product of abelian surfaces.

The proof chain is:

$$\underbrace{\text{palindromic } g(x)}_{\text{input}} \xrightarrow{\mu=(1/x,y/x^5)} \underbrace{D_8\text{-action}}_{\text{group theory}} \xrightarrow{\text{Kani–Rosen}} \underbrace{J(C_t) \sim A^2}_{\text{splitting}} \xrightarrow{\text{Ribet}} \underbrace{\omega \text{ algebraic}}_{\text{Hodge Conj.}}$$

This is the ninth CRMLint Squeeze in the series. The constructive content (BISH) identifies the geometric structure; the classical content (CLASS) provides the Hodge-theoretic conclusion.

Two directions remain open: extending the algebraicity result to the Paper 85 family (which has no reciprocal involution), and finding a structural explanation for the universal vanishing $\tau_+ = 0$.

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Lean 4 source and CAS scripts: included in the Zenodo archive.

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